ELECTRON TRANSPORT CHAIN AND OXIDATIVE PHOSPHORYLATION

ATP as an energy carrier

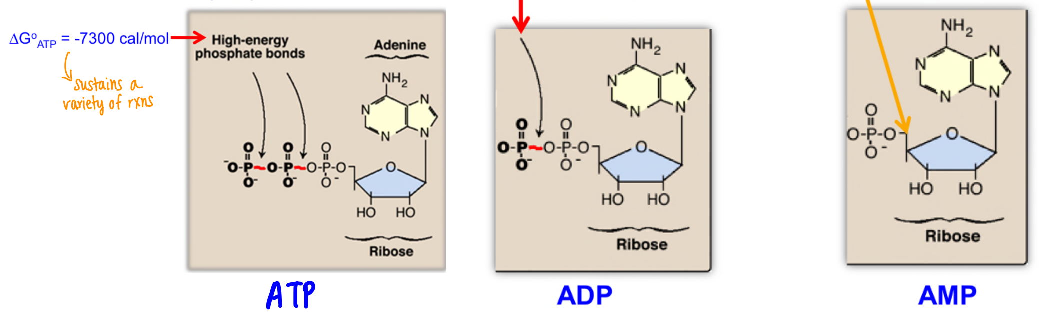
- ATP = adenosine + 3 phosphates

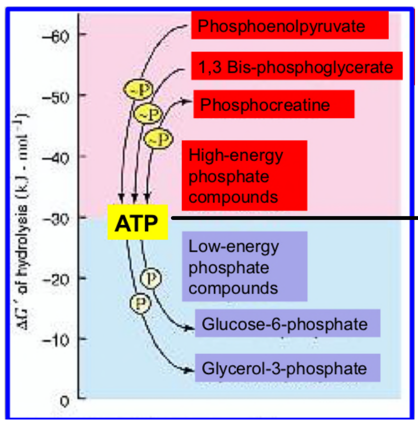
ΔG^o_{ATP} = -7300 cal/mol → sustains a variety of rxns

ΔG^o_{ADP} = -7300 cal/mol

High energy bond

ΔG^o < -4000 cal/mole



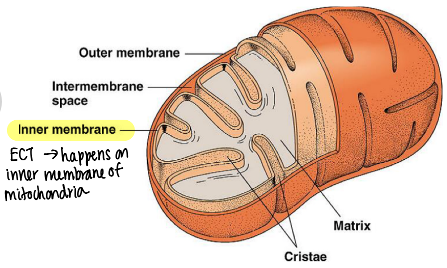


very high energy phosphate compounds

ATP = somewhere in the middle

low-energy phosphate compounds

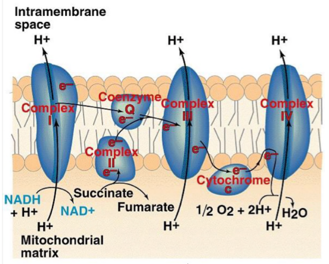
Electron Transport Chain & oxidative phosphorylation



ECT → happens on inner membrane of mitochondria

Function
- oxidize NADH + $FADH_2$
- generate electrical energy by passing electrons to oxygen
- create a proton gradient across inner mitochondrial membrane.
- proton gradient drives phosphorylation of ADP to ATP

Reactions of the ETC

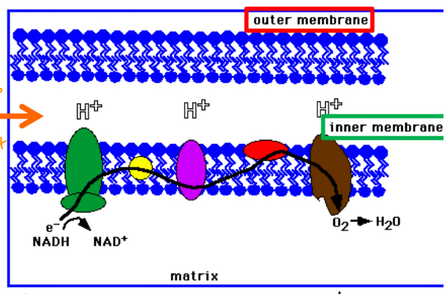


1. Electron Transport
2. oxidative phosphorylation + the chemiosmotic hypothesis

Electron Transport results when NADH & $FADH_2$ donate electrons to complexes in the inner mitochondrial membrane. The electrons flow through the complexes and are eventually donated to oxygen forming water.

Intermembranous space

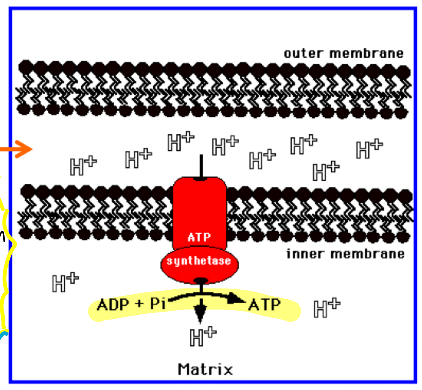
→ proton gradient is created



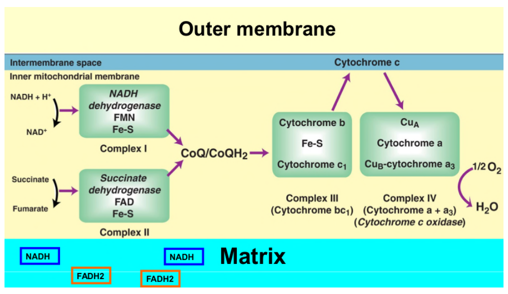
The inner membrane of mitochondria contains the proteins of the electron transport chain, and is the barrier allowing the formation of a H^+ gradient for ATP production through ATP synthase.

intermembrane space

This process pumps protons (H^+) into the intermembrane space. The gradient created (high concentration of protons in the intermembrane space and low concentration in the matrix) causes protons to flow through ATP synthase in the inner membrane resulting in production of ATP.



Electron Transport chain Overview



① Electron Transport

Involves four protein complexes:
- Complex I = NADH dehydrogenase
- Complex II = Succinate dehydrogenase
- Complex III = Cytochrome reductase
- Complex IV = Cytochrome oxidase

☆ Complexes I – IV accept/donate electrons to relatively mobile electron carriers such as:
- Coenzyme Q
- Cytochrome C

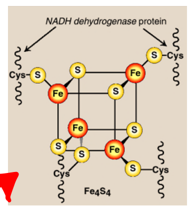
Prosthetic groups in each complex reversibly accept + release electrons

- Complex I – FMN
- Complex II – FAD

} riboflavin derivative

- Complex III – heme group (Fe^{3+}) → iron is ferric
- Complex IV – Cu^{2+} and heme group (Fe^{3+})

Complexes I, II, and III also contain iron-sulfur proteins.



Where the electrons come from:

Complex I: NADH electrons come from...
1. Malate dehydrogenase (MDH)
2. α-ketoglutarate dehydrogenase (α-KGDH)
3. isocitrate dehydrogenase (IDH)
4. Pyruvate Dehydrogenase (PDH)

+ from fatty acid β-oxidation + from cystolic sources such as glycolysis.

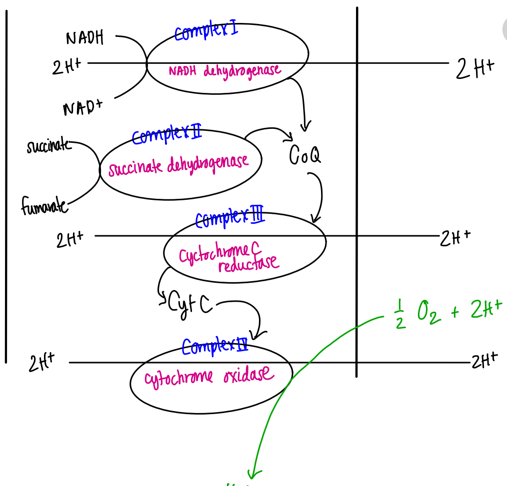
Complex II: $FADH_2$ → made in TCA cycle + fatty acid synthesis.
- $FADH_2$ electrons from succinate dehydrogenase or associated CoQ gets them from glycerol phosphate shuttle or from fatty acid β-oxidation.

Flow of electrons

1. NADH is oxidized by CoQ at complex I and/or $FADH_2$ is oxidized by CoQ at complex II
2. $CoQH_2$ is oxidized by cytochrome C at complex III
3. Cytochrome C is oxidized by O_2 at complex IV

O_2 = final electron acceptor at complex IV!

mitochondrial matrix



intramembrane space

proton gradient is being formed.

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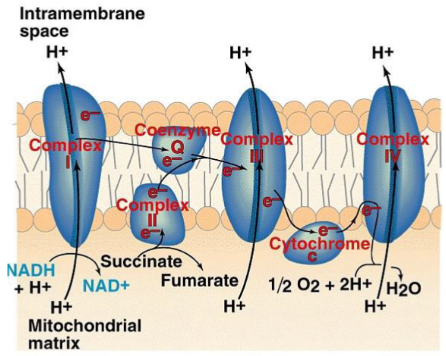
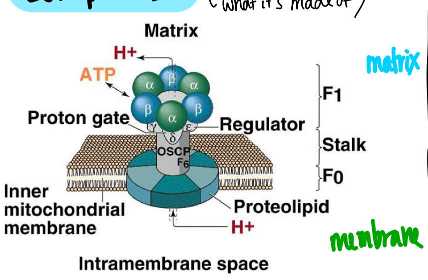


② oxidative phosphorylation (and the chemiosmotic Hypothesis)

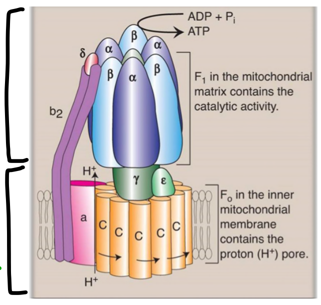
- As electrons flow down electrochemical potential, protons are pumped into the intermembrane space.
- Protons are pumped into intramembrane space at complexes I, III, and IV.
- This creates a pH gradient that is relieved by pumping protons back through F_0F_1-ATP synthase (complex V). The energy released in this process is coupled to ATP synthesis from ADP + P_i.

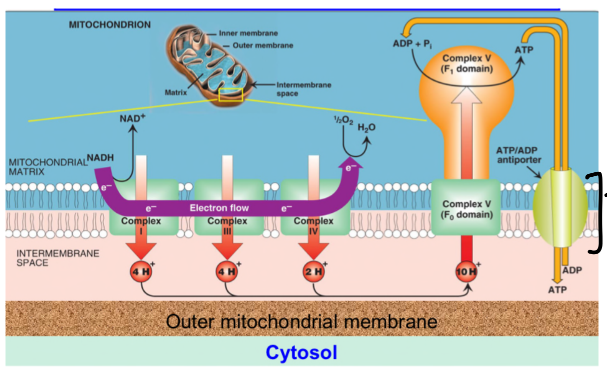
Complex V: (don't need to know what it's made of)



matrix

membrane





transports ATP out of membrane

Inhibition of Electron Transport

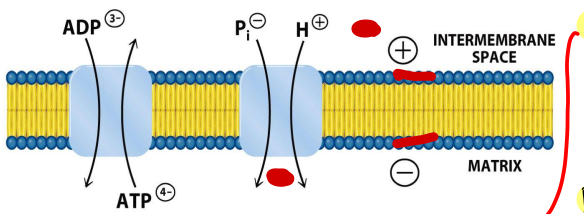
- Rotenone, piericidin A (bacterial antibiotic) (pesticide) + amytal (barbiturate) — Inhibit NADH dehydrogenase in complex I
- Amtimycin A (antibiotic) — Inhibits cytochrome b of cytochrome reductase in complex III
- CO, sodium azide, hydrogen sulphide (H_2S) and cyanide (CN^-) → inhibit cytochrome oxidase (complex IV)

- Oligomycin (streptomyces antibiotic) — inhibits ATP synthase (complex V)

Inhibitors decrease ATP synthesis, decrease ETC, and oxygen consumption.

Transporting ATP + ADP

- Adenine nucleotide translocase: unidirectional exchange of ATP for ADP (antiport)
- symport of P_i and H^+ is electroneutral



effect of both toxins are similar to oligomycin

Inhibiting ATP/ADP Transport

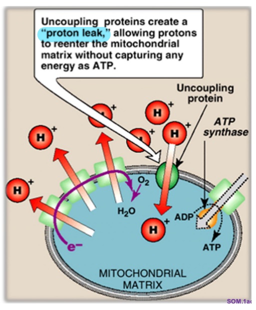
Atractyloside: toxic glycoside (molecule w/ sugar + noncarbohydrate element) from thistle plant *Atractylis gummifera*
↳ Binds the outward facing (inter-mem. space) portion of adenine nucleotide transporter.

Bongkrekic Acid: respiratory toxin produced in coconuts contaminated with *Burkholderia gladioli*
↳ Binds the inward facing (matrix) portion of adenine nucleotide transporter

Uncouplers of oxidative phosphorylation

- dinitrophenol (DNP), Aspirin, thermogenin, ionophores act by destroying the proton gradient

uncouplers decrease ATP synthesis, increase ETC, and oxygen consumption
They basically dissipate the proton gradient via punching holes in the inner mitochondria.



- **Thermogenin** : "uncoupling protein" → leads to leaky membranes

Dissipation of the H^+ gradient generated from electron transport, which is uncoupled from ATP synthesis, generates heat.

↳ How the physiological function of brown fat works.
↳ Thermogenin = brown b/c of the abundance of cytochrome-containing mitochondria.
↳ newborns have brown fat in neck + ↑er back = biological heating pad.
↳ hormone-induced release of fatty acids from triglycerides in brown fat leads to activation of thermogenin.

- **Ionophores** : increase the permeability of the inner mitochondrial membrane and uncouples electron transport from oxidative phosphorylation.
  - → make the inner membrane permeable to compounds that cannot usually cross.
  - → can transport different kinds of ions whereas uncouplers specifically increase proton permeability.
  - ↳ can be channel formers or mobile carriers

ie) Gramicidin → channel-forming ionophore
→ mixture of abx produced by Bacillus brevis

Valinomycin → mobile carrier
→ Abx from streptomyces strains
→ typically associated with carrying K^+ across bilayers
→ polar inside for the proton & non-polar/lipophilic outside